

$$\text{var}(j_1^T Y) = \text{var}(j_2^T Y) \Rightarrow \text{var}((j_1 - j_2)^T Y)$$

Case I: $\text{var}(j_1^T Y) > \text{var}(j_2^T Y) \rightarrow \text{contradiction}$

Case II: $\text{var}((j_1 - j_2)^T Y) = 0$
 $\Rightarrow E((j_1 - j_2)^T Y) = 0$

$$P((j_1 - j_2)^T Y = 0) = 1$$

$$P(j_1^T Y = j_2^T Y) = 1$$

LINEAR MODELS

- Simple Linear
- Multiple Linear
- Polynomial

Assumption

① $Y \sim N(X\beta, \sigma^2 I_n)$

② $(X^T X)$ is invertible

$$Y = X\beta + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\text{If } |X^T X| \approx 0$$

$$\hat{\beta} = \frac{\text{adj}(X^T X)}{|X^T X|} \cdot X^T Y$$

Teacher's Signature _____

Impact

- ① Some of the β_i will take large values
- ② $\text{var}(\hat{\beta}_i)$ will be large $\rightarrow$ some large, some small values
- ③ $\hat{y}_i = x_i^T \hat{\beta}$ will have large error

Why would such a scenario occur?

any Multicollinearity

- ① Improper sample survey
- ② If we consider higher degree of proportion vector

Consider

$$E[(\beta - \hat{\beta})^T C (\beta - \hat{\beta})]$$

$$= E\left(\sum_{i=0}^k (\beta_i - \hat{\beta}_i)^2\right)$$

$$= \sum_{i=0}^k E[(\beta_i - \hat{\beta}_i)^2]$$

We know $E(\hat{\beta}_i) = \beta_i$

$$= \sum_{i=0}^k \text{var}(\hat{\beta}_i)$$

$\rightarrow$ var $\rightarrow$ will be diagonal elements & trace

We know $\hat{\beta} \sim N(\beta, \sigma^2 C(X^T X)^{-1})$

$$= \sigma^2 \sum_{i=0}^k C_{ii} = \boxed{\sigma^2 \text{tr}(C)}$$

Teacher's Signature

Note

- ① $X^T X$ is a symm matrix, here p.s.d
- ② $(X^T X)^{-1}$ is also a symm matrix
- ③ Let the eigenvalues of $(X^T X)$ be $\lambda_0 > \lambda_1 > \dots > \lambda_k > 0$
- ④ $\lambda_{\max}^{(A)} = \max_{\|x\|=1} x^T A x$
- $\lambda_{\min}^{(A)} = \min_{\|x\|=1} x^T A x$

where A is a p.s.d symmetric matrix

Hence

$$\text{tr}(X^T X) = \sum_{i=0}^k \lambda_i$$

$$\text{tr}(X^T X)^{-1} = \sum_{i=0}^k \frac{1}{\lambda_i} \quad (\text{if } \lambda_i > 0)$$

$$|X^T X| = \prod_{i=0}^k \lambda_i$$

$$|(X^T X)^{-1}| = \prod_{i=0}^k \frac{1}{\lambda_i} \quad \text{if } \lambda_i > 0$$

HENCE

$$E[(\beta_i - \hat{\beta}_i)^2] = \sigma^2 \text{tr}(C)$$

Teacher's Signature

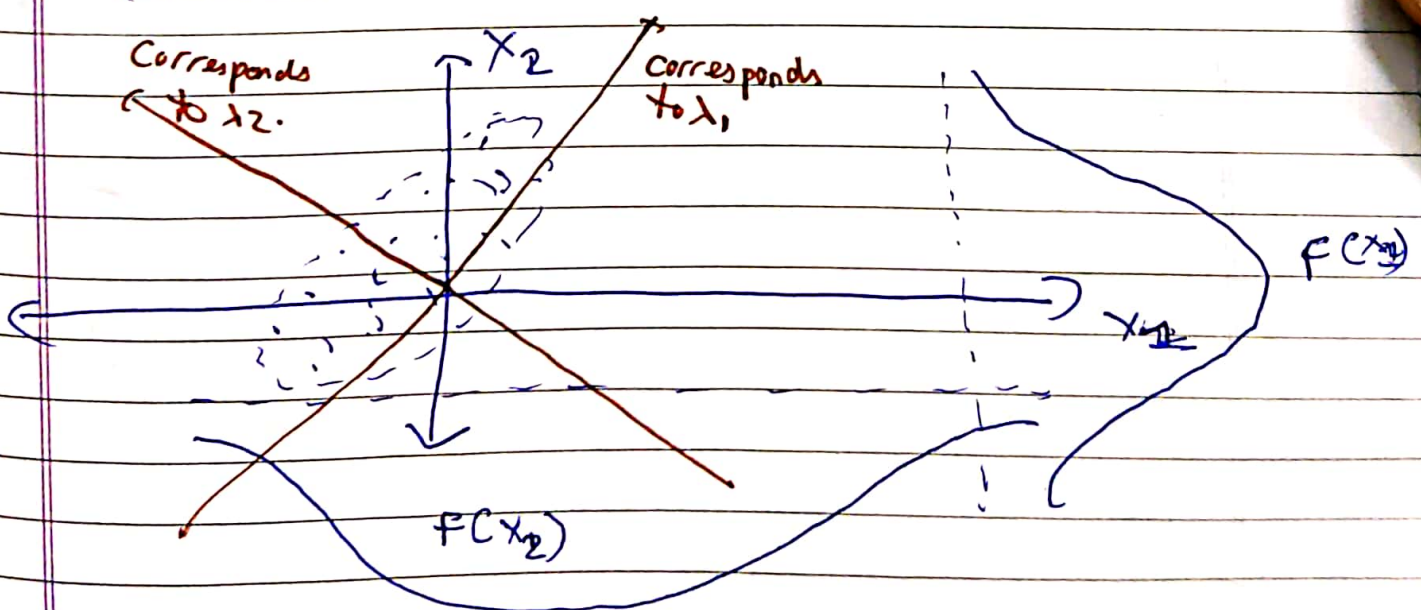
$$= \sigma^2 \sum_{i=0}^K \frac{1}{\lambda_i}$$

We can clearly see that if $\lambda_i \rightarrow 0$ then $E(\sum (\beta_i - \hat{\beta}_i)^2) \rightarrow \infty$. This gives you an indication of how many λ_i you can incorporate in your regression.

Usually $|X^T X| \approx 0$ is called as a Multicollinearity problem. The possible solutions are:-

- ① Principal Component Regression
 - ② RIDGE $\rightarrow [\|\hat{\beta}\|_2 < C_2]$
 - ③ LASSO $\rightarrow [\|\hat{\beta}\|_1 < C_1]$
- restricting the domain of $\hat{\beta}$ vector to prevent its components from blowing up.

* PRINCIPAL COMPONENT ANALYSIS



Teacher's Signature _____

① PCA of X will give a data driven co-ordinate system

② will give orthogonal ~~co-ordinate~~ co-ordinates of the system.

$$(X^T X)$$

$$\lambda_0 > \lambda_1 > \lambda_2 \dots > \lambda_K \rightarrow \text{eigenvalues}$$

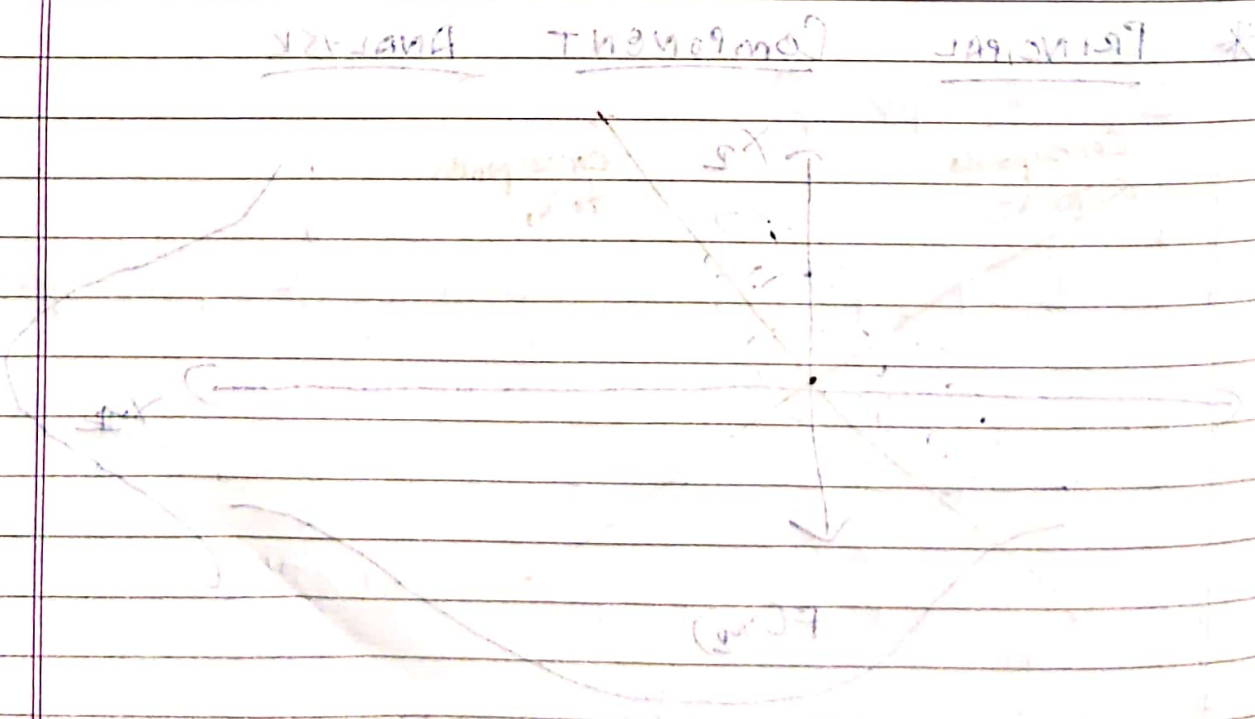
$$u_0 > u_1 > u_2 \dots > u_K$$

$$(X^T X) u = \lambda u$$

Solve $(X^T X - \lambda I) = 0$ to obtain λ_i

$\Rightarrow Z = (u_0, u_1, \dots, u_K)$
 $\searrow$
 Score $\rightarrow$ co-ordinate in the new system.

To obtain a regression between Y and Z_i .



Teacher's Signature _____

PCR

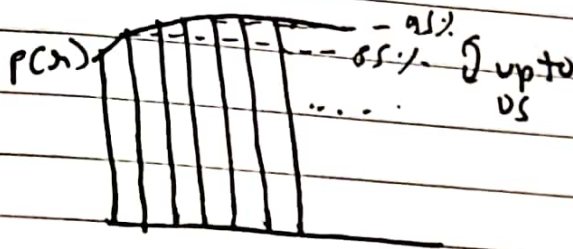
$$\lambda_0 > \lambda_1 > \dots > \lambda_k > 0 \rightarrow \text{eigenvalues}$$

$$u_0, u_1, \dots, u_k \rightarrow \text{eigen vectors}$$

Variation along i th eigen direction $\frac{u_i}{\|u_i\|}$ is λ_i

$$\text{Total variation} = \sum_{i=0}^k \lambda_i$$

Proportion of variation explained $p(k) = \frac{\sum_{i=0}^k \lambda_i}{\sum_{i=0}^{\infty} \lambda_i} \quad 0 \leq k$



$$P = [u_0, u_1, \dots, u_k]$$

$(k+1, k+1)$

$$X^T X = P D P^T$$

$(k+1 \times k+1)$

$$D = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_k)$$

Singular value decomposition
or PCA

$$P P^T = P^T P = I$$

$\hookrightarrow$ orthogonal matrix

- We will do a transformation of X to Z through P and we will use the first k components as regressors. $\rightarrow$ Dimensionality Reduction.

$$Y = X\beta + \epsilon$$

$$= X I_{k+1} \beta + \epsilon$$

Teacher's Signature

$$= X P P^T \beta + \epsilon$$

~~α~~

$$\approx \underbrace{X P_{(h+1)}}_Z \underbrace{P_{(h+1)}^T}_T \beta + \epsilon$$

↳ first $h+1$ columns of P
 $C (h+1 \times h+1)$
 $h+1$

$$= Z \gamma + \epsilon$$

$$\hat{\alpha}_L = (Z^T Z)^{-1} Z^T \gamma$$

$(h+1 \times h+1)$

HW:- Check if $(Z^T Z)^{-1}$ is diagonal

Now

$$\beta \approx I \beta$$

$$= P P^T \beta$$

$$\hat{\beta}_{PC} \approx P \begin{pmatrix} \hat{\alpha} \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$h+1$
 x_{h+1}
 $h+1$

$$\hat{\alpha} \in \mathbb{R}^{h+1}$$

$$x \in \mathbb{R}^{k+1}$$

$$Z \sim \mathbb{R}^{n+1}$$

$$n < k$$

$$\hat{y} = (\hat{\beta}_{PC})^T x$$

Now for any new x , instead of converting it into a z every time, we can use $\hat{\beta}_{PC}$ directly.

* SHRINKAGE METHOD / REGULARIZATION

Multi collinearity $\Rightarrow |X^T X| \approx 0$

Teacher's Signature

$\Rightarrow \hat{\beta}_{LS}$ may have a large variance. We want to have bounds on the length of β

↓
Restricted Optimization Problem

$\hat{\beta} = \underset{\|\beta\| < c}{\operatorname{argmin}} (Y - X\beta)^T C (Y - X\beta) \rightarrow \text{L.S. condition with a bound on } |\beta|$

error matrix

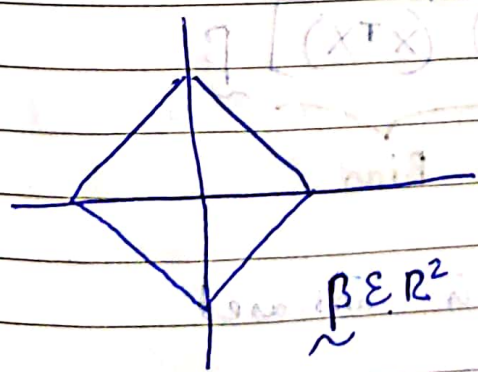
- L_1 norm
 $L_1(Z) = \sum_{i=1}^n |z_i| \rightarrow \text{LASSO}$

- L_2 norm
 $L_2(Z) = \left(\sum_{i=1}^n z_i^2 \right)^{1/2} \rightarrow \text{RIDGE}$

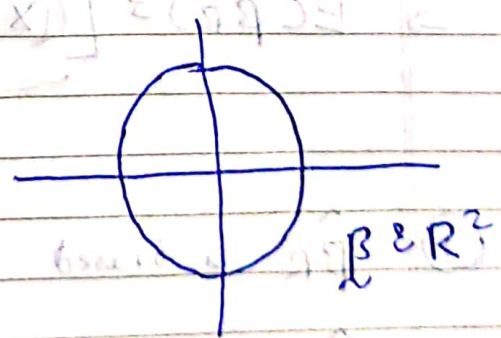
- $L_\infty(Z) = \max_{1 \leq i \leq n} |z_i| \rightarrow ?$

- If we assume $\beta \sim N(\mu_\beta, \Sigma_\beta)$

then we can use
 $(\beta - \mu_\beta)^T \Sigma_\beta^{-1} (\beta - \mu_\beta)$



$\|\beta\|_1 < c_1$



$\|\beta\|_2 < c_2$

Teacher's Signature

In RIDGE regression we minimised LS condition
wrt. $\sum_{i=0}^k \beta_i^2 < C$

ie we minimised $SC(\beta) = (Y - X\beta)^T(Y - X\beta)$ wrt
 $\sum_{i=0}^k \beta_i^2 < C$

$$\hat{\beta}_R = \underset{\beta}{\operatorname{argmin}} SC(\beta) + \lambda C\beta^T\beta - C$$

$\swarrow$ on solving $\lambda > 0$ $\nwarrow$ Lagrange multipliers

$$\Rightarrow \hat{\beta}_R = (X^T X + \lambda I)^{-1} X^T Y \rightarrow \text{LEARN!!!}$$

Ridge estimate of regression coefficients.

- Due to λ , we are incorporating some amount of bias in the system. If $\lambda \rightarrow 0$ $\hat{\beta}_R = \hat{\beta}_{LS}$ and if $\lambda \rightarrow \infty$ $\hat{\beta}_R = 0$.
 $\downarrow$ unbiased

$$\hat{\beta}_R = (X^T X + \lambda I)^{-1} X^T Y$$

$$= [(X^T X + \lambda I)^{-1} (X^T X)] [C X^T X]^{-1} X^T Y$$

$\downarrow \hat{\beta}_{LS}$

$$\Rightarrow E(\hat{\beta}_R) = [(X^T X + \lambda I)^{-1} (X^T X)] \beta$$

Bias $\rightarrow$ due to putting a bound of $\|\beta\| < C$

① $\hat{\beta}_R$ is biased but $\hat{\beta}_{LS}$ is unbiased $\|\beta\| < C$

② $\hat{\beta}_R$ has less dispersion compared to $\hat{\beta}_{LS}$



Teacher's Signature _____

* ERROR ANALYSIS

True error $\epsilon \sim N(0, \sigma^2 I_n)$

Estimated error $e = y - \hat{y} \sim N(0, \sigma^2 (I_n - P_x))$

① e_i are dependent

② $E(e_i) = 0$

③ $\text{var}(e_i) = \sigma^2 (1 - h_{ii})$ not id

④ $e_i \sim N(0, \sigma^2 (1 - h_{ii}))$ marginally

$$\text{cov}(e_i, e_j) = \begin{cases} (1 - h_{ii}) \sigma^2 & i=j \\ (-h_{ij}) \sigma^2 & i \neq j \end{cases}$$

$$\hat{\sigma}^2 = \frac{SSE}{n-k-1} \quad \xrightarrow{\text{SSR}} \sum (y_i - \hat{y}_i)^2$$

$$g_i = \frac{e_i}{\sqrt{(1 - h_{ii}) \sigma^2}} \sim N(0, 1)$$

$$\hat{g}_i = \frac{e_i}{\sqrt{(1 - h_{ii}) \hat{\sigma}^2}} \sim t_{n-k-1}$$

- If n is large compared to k

$$\hat{\sigma}^2 \rightarrow \sigma^2 \text{ with prob. 1}$$

and $\hat{g}_i \sim N(0, 1)$ approximate

$$\hat{g}_i \xrightarrow{d} N(0, 1)$$

Converges in distribution $\rightarrow$ Google $\rightarrow$ SLLN
Almost sure,

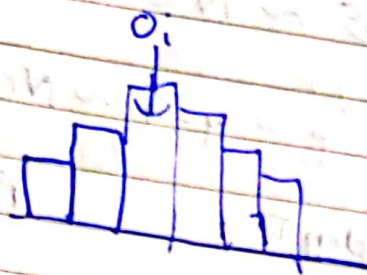
$\downarrow$
L₂ convergence in probability,
Convergence in distribution
WLLN

Teacher's Signature $\rightarrow$ Binomial to Poisson (same cdf)

To check normality

(i)
(ii)

QQ plot
 χ^2 plot

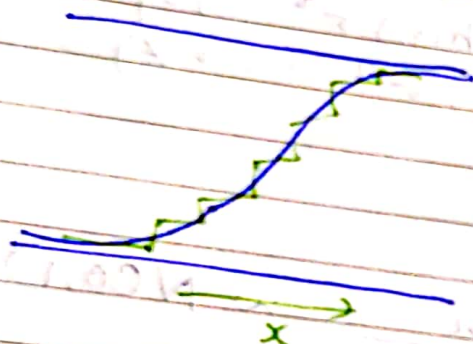


$$E_i = n \int_{a_i}^{b_i} f(x) dx$$

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1}$$

(iii)

Kolmogorov Smirnov test



$$F_n(x) = \sum_{i=1}^n \frac{I_{x_i \leq x}}{n}$$

$$\max_x \sqrt{n} |F_n(x) - F(x)| \text{ is the test statistic}$$

$$\hookrightarrow \sup_t (|B_0(t)|)$$

$B_0(t)$ is the standard Brownian Bridge on $[0, 1]$

OUTLIER DETECTION

- Leave one out or Jack-knife method.
To check that the dataset has outliers or not.

Teacher's Signature

$$\sum_{i=1}^n (y_i - \hat{y}_{ci})^2 = \text{PRESS}$$

$\hat{y}_{ci}$ is the predicted value of y_i when (x_i, y_i) has not been used for estimation of ~~parameter~~ β .

$$e_{ci} = y_i - \hat{y}_{ci} \equiv \frac{e_i}{1 - h_{ii}} \sim N\left(0, \frac{\sigma^2}{1 - h_{ii}}\right)$$

$$\frac{e_{ci}}{\sqrt{\hat{\sigma}_{ci}^2 / (1 - h_{ii})}} = \frac{e_i}{\sqrt{\sigma_{ci}^2 (1 - h_{ii})}}$$

$$S_i^2 = \hat{\sigma}_{ci}^2 = \frac{(n - k - 1) \text{MSE}_{\text{error}} - \left(\frac{e_i^2}{1 - h_{ii}}\right)}{n - k - 2}$$

$$t_i = \frac{e_i}{\sqrt{S_{ci}^2 (1 - h_{ii})}} \sim t_{n-k-1} \text{ under } H_0$$

H_0 : i th data is NOT an outlier
 H_1 : i th data is an outlier

Teacher's Signature

* TIME SERIES

- Collection of r.v. $\{X_t / t \in T\}$ over a time index set T , which might be finite, countably infinite or uncountably.
- Realised value: What we observe are the realised values of time series ie the data set is $\{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\}$ where x_i s are some numeric or categorical values.
- Time is governed by the solar and lunar cycle.
- Discrete Time Series: $T \rightarrow$ countable set $\Rightarrow$ discrete
- Continuous Time Series: $T \rightarrow$ interval $\Rightarrow$ continuous.
- Discrete & continuous are the adjectives of time, NOT of the r.v. X_t .

Eg)

$$\begin{aligned} 1. \quad X_t &= +1 \rightarrow p \\ &= -1 \rightarrow 1-p \end{aligned} \quad \} \text{Gambler's ruin}$$

$$2. \quad X_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$3. \quad S_t = \sum_{i=0}^t X_i$$

↓
Random Walk

$$X_t = \begin{cases} +1 & p \\ -1 & 1-p \end{cases}$$

$X_0 = 0$

Note: If $p = 1/2$ then you will eventually come back to the origin with prob=1.

This works on $\mathbb{Z}^2$ as well, but it will not work for $\mathbb{Z}^d, d \geq 3$. (You can find this in Norris, Markov chains)

4. $X_t \stackrel{iid}{\sim} N(0, \sigma^2)$

$$S_t = \sum_{i=0}^t X_i$$

$$E(S_t) = 0$$

$$\text{var}(S_t) = t\sigma^2$$

Now if $X_t \stackrel{iid}{\sim} N(0, \sigma^2) + \delta$

$$E(S_t) = \delta$$

$$\text{var}(S_t) = t\sigma^2$$

- If the distribution does not change $\Rightarrow$ Strong Stationarity

- If the mean and variance does not change $\Rightarrow$ Weak Stationarity

$\therefore X_t \rightarrow$ strongly stationary

$S_t \rightarrow$ non stationary

Now if we make

$$S^*_t = \frac{1}{\sqrt{t}} \sum_{i=0}^t (X_i - \delta)$$

$$E(S^*_t) = 0$$

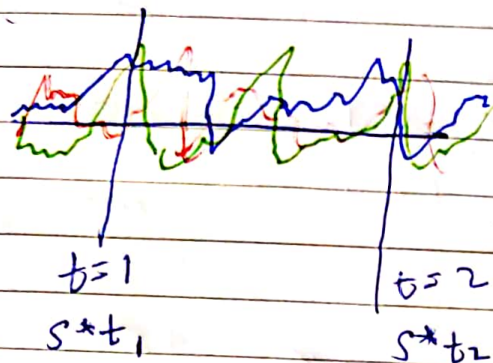
$$\text{var}(S^*_t) = \sigma^2$$

weakly stationary

Teacher's Signature

* Distribution of Time Series

- All such realisations are at particular time instant.



(Y_{t1}, Y_{t2}) may be uncorrelated or correlated or independent. Any general time series

$\{W_t\}$ with $E(W_t) = 0$

$$\text{var}(W_t) = \sigma^2$$

$$\text{corr}(W_t, W_s) = 0$$

Then $\{W_t\}$ is called White Noise

$W_t \rightarrow$ Mean 0

Variance σ^2

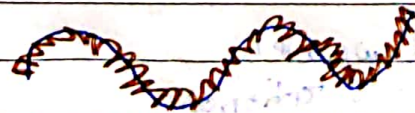
iid

* **WN:** $W_t \sim WN(0, \sigma_w^2)$

$\hookrightarrow$ WN need not be normally distributed

Ex SNR - Signal to Noise Ratio:

$$X_t = A \sin(2\pi f t + \phi) + W_t \quad \text{R.V. at timestamp.}$$



Otherwise collection of R.V.

Teacher's Signature _____

Ex- Moving average process - Corder 1) MAC4)

$$W_t \sim WN(0, \sigma_w^2)$$

$$X_t = W_t + \theta W_{t-1}$$

$$= 1 \cdot W_t + \theta W_{t-1} + 0 \cdot W_{t-2} + 0 \cdot W_{t-3} + \dots$$

The process is itself constructed through noise.

→ Autoregressive Process :- Corder = 1) ARC1)

$$W_t \sim WN(0, \sigma_w^2)$$

$$X_t = \phi X_{t-1} + W_t \Rightarrow y = L(X) + \text{error}$$

$$= \phi (\phi X_{t-2} + W_{t-1}) + W_t$$

$$= W_t + \phi W_{t-1} + \phi^2 X_{t-2} + \dots$$

- ARC1) process is same as MAC(∞) process
 $|\phi| < 1$

Regression with itself → Autoregression.

→ Brownian Motion → Lesson

$\{X_t\}$ is said to follow Brownian Motion if

→ Initial point fixed

① $X_0 = a$ (if $a=0 \rightarrow$ Standard Brownian motion)

② $X_{t+s} - X_t$ is independent of t

③ $X_{t+s} - X_t \sim N(0, s)$

④ $\{X_t\}$ has continuous path but not differentiable

Teacher's Signature _____

$$W_k \stackrel{iid}{\sim} N(0, 1)$$

$$X_t = \frac{1}{\sqrt{n}} \sum_{k=0}^{[nt]} W_k$$

$n \rightarrow \infty \quad t \in (0, 1)$

Simulation of Brownian Motion

$$X_t = \frac{\sqrt{[nt]}}{\sqrt{n}} \left(\frac{1}{\sqrt{[nt]}} \sum_{k=1}^{[nt]} W_k \right)$$

Normalization
like $S_t^* = \frac{1}{\sqrt{t}} \sum_{i=0}^t (X_i - \bar{X})$

↓ as $n \rightarrow \infty$

$$X_t \sim \sqrt{t} N(0, 1) = N(0, t)$$

$$E(S_t^*) = 0$$

$$\text{Var}(S_t^*) = \sigma^2$$

Now we will ~~consider~~ construct a vector of X_t for observations in different timestamps.

$$0 < t_1 < t_2 < \dots < t_k < \dots$$

$$(X_{t_1}, X_{t_2}, \dots, X_{t_k}) \sim N \left[0, \begin{pmatrix} t_1 & \min(t_1, t_2) & \dots & \min(t_1, t_k) \\ t_2 & & \ddots & \\ \vdots & & & \\ t_k & & & \end{pmatrix} \right]$$

if $t < s$

$$\begin{pmatrix} X_t \\ X_s \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} t & t \\ t & s \end{pmatrix} \right)$$

Why is $\text{cov}(i, j) = \min(t_i, t_j)$

$$X_t = \frac{1}{\sqrt{n}} \sum_{k=1}^{[nt]} W_k$$

only the min term will be taken into account

$$X_s = \frac{1}{\sqrt{n}} \sum_{k=1}^{[ns]} W_k = \frac{1}{\sqrt{n}} \left(\sum_{k=1}^{[nt]} W_k + \sum_{k=[nt]+1}^{[ns]} W_k \right)$$

* BROWNIAN BRIDGE

Consider $\{X_t\}$ to be a B.M. conditioned to $X_0 = 0$ and $X_1 = 0$, $t \in [0, 1]$, then

X_t is said to follow standard Brownian Bridge with the following distribution property:-

$$B_t \sim B \text{ Bridge } [0, 1]$$

$$\begin{pmatrix} B_t \\ B_s \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} t(1-t) & t(1-s) \\ t(1-s) & s(1-s) \end{pmatrix} \right)$$

$$B(t) = W(t) - t \cdot W(1)$$

Brownian Motion

$$W(t=1) = W(1) \\ \text{Hence } B(1) = 0$$

$$\begin{aligned} \text{var}(B(t)) &= \text{var}(W(t) - t \cdot W(1)) \\ &= \text{var}(W(t)) + \text{var}(t \cdot W(1)) - 2\text{cov}(W(t), t \cdot W(1)) \\ &= t + t^2 - 2t^2 \\ &= t^2 + t = t(1+t) \end{aligned}$$

* K.S. Test

$$\text{Under } H_0: K = \max_{0 \leq t \leq 1} |B(t)|$$

$$B_t \sim B \text{ Bridge } [0, 1]$$

Use $\rightarrow X_1, \dots, X_n$ iid const

$H_0: X \sim F_0$ (X follows F_0 distribution)

$H_1: X$ doesn't follow F_0

If H_0 is true $\Leftrightarrow$ If $X \sim F_0(\cdot)$

$$T = F_0(X) \sim ?$$

Teacher's Signature

$$\begin{aligned}
 P(C \leq t) &= P(F_0(X) \leq t) \\
 &= P(X \leq F_0^{-1}(t)) \quad \text{by defn of CDF} \\
 &= P_0(F_0^{-1}(t)) = t
 \end{aligned}$$

$$T = F_0(t) \sim U[0,1] \text{ under } H_0$$

$$T_i = F_0(x_i)$$

$$H_0: T_i \sim U[0,1]$$

vs

$$H_1: T_i \text{ does not follow } U[0,1]$$

$$G_n(t) = \frac{1}{n} \sum_{i=1}^n I\{T_i \leq t\}$$

I: indicator

each indicator I is Bernoulli

with success prob t

Its summation will be binomial

$$E\left(\frac{1}{n} \sum_{i=1}^n I\{T_i \leq t\}\right) = \frac{nt}{n} = t$$

$$\text{var}\left(\frac{1}{n} \sum_{i=1}^n I\{T_i \leq t\}\right) = \frac{n t(1-t)}{n^2} = \frac{t(1-t)}{n}$$

$$E(G_n(t)) = t$$

$$\text{var}(G_n(t)) = \frac{t(1-t)}{n}$$

Test Stat for Hypothesis Testing :-

$$\sup_t \sqrt{n} |G_n(t) - t| \xrightarrow{d} K = \sup_{0 \leq t \leq 1} |B_0(t)|$$

$$\sqrt{n} (G_n(t) - t)$$

$$\xrightarrow{d} N(0, t(1-t))$$

as $n \rightarrow \infty$ CDF will overlap with $N(0, t(1-t)) \Rightarrow$ convergence

in dist.

normally distributed at a time step.

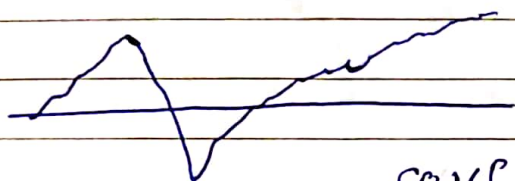
- While computing the test stat, replace $\sup_t$ with max. over the func $|G_n(t) - t|$

MEAN & AUTO COVARIANCE

- Suppose that $\{X_t\}$ is a time series with $E|X_t| < \infty$. Its mean function is $\mu_t = E[X_t]$

- If $E[X_t^2] < \infty$, then its autocovariance function $\gamma_{xx}(s, t)$ is

$$\gamma_{xx}(s, t) = \text{cov}(X_s, X_t) = E[(X_s - \mu_s)(X_t - \mu_t)]$$



$$\text{cov}(X_s, X_{s+1}) = \gamma_{xx}(s, s+1)$$

X_1	
X_2	X_2
$\vdots$	$\vdots$
X_n	X_{n-1}
	X_n

Paired $n-1$ data points to calculate the covariance.

Sample
ACVF



$$\gamma_{xx}(s, s+1) \hat{=} \left(\frac{1}{n} \right) \sum_{i=1}^{n-1} (X_i - \bar{X})(X_{i+1} - \bar{X})$$

$\downarrow$ lag of 1 $\downarrow$ always $\bar{X}$ $\downarrow$ always $\bar{X}$

$$\gamma_{xx}(s, s+h) = \frac{1}{n} \sum_{i=1}^{n-|h|} (X_i - \bar{X})(X_{i+h} - \bar{X})$$

If n is changed to $n-h$, then the estimated autocovariance matrix may not be p.s.d.

Teacher's Signature _____

Accordingly, if a time series is stationary, the mean function (1.21) $\mu_t = \mu$ is constant so that we can estimate it by the sample mean,

$$\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t. \quad (1.32)$$

The standard error of the estimate is the square root of $\text{var}(\bar{x})$, which can be computed using first principles (recall footnote 3 on page 20), and is given by

$$\begin{aligned} \text{var}(\bar{x}) &= \text{var} \left(\frac{1}{n} \sum_{t=1}^n x_t \right) = \frac{1}{n^2} \text{cov} \left(\sum_{t=1}^n x_t, \sum_{s=1}^n x_s \right) \\ &= \frac{1}{n^2} \left(n\gamma_x(0) + (n-1)\gamma_x(1) + (n-2)\gamma_x(2) + \cdots + \gamma_x(n-1) \right. \\ &\quad \left. + (n-1)\gamma_x(-1) + (n-2)\gamma_x(-2) + \cdots + \gamma_x(1-n) \right) \\ &= \frac{1}{n} \sum_{h=-n}^n \left(1 - \frac{|h|}{n} \right) \gamma_x(h). \end{aligned} \quad (1.33)$$

autocovariance matrix

$$P(X_1 \leq x_1)$$

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$$\Gamma = (\gamma_{i,j})$$

$$= (\gamma(x_i, x_j)) = (\gamma_{i-j})$$

$\Downarrow$

	X_1	X_2	...	X_n	
X_1	$\gamma(0)$	$\gamma(1)$	...	$\gamma(n-1)$	
X_2	$\gamma(1)$	$\gamma(0)$	$\gamma(1)$	...	$\gamma(n-2)$
$\vdots$	$\gamma(2)$	$\gamma(1)$	$\gamma(0)$	...	$\gamma(n-3)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_n	$\gamma(n-1)$	$\gamma(n-2)$	...	$\gamma(0)$	

- A time series $\{X_t\}$ is said to be ~~weakly~~ weakly stationary if

1. μ_t is independent of t .
2. For each $h \in \mathbb{Z}$ ACVF $\gamma(t+h, t)$ is independent of t .

Example

$$X_t \sim WN(0, \sigma_w^2)$$

- $\{X_t\}$ is strongly stationary if $\forall k, h$
 $t_1, t_2, \dots, t_k$ shifting the time axis does not affect the distribution. i.e.

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k)$$

Example

$$X_t \sim WN = N(0, \sigma^2)$$

Strong Stationary $\Rightarrow$ weak stationarity

Teacher's Signature

Imp

 $\{X_t\}$ iid $N(0, \sigma^2)$

$$S_t = \sum_{i=1}^t X_i \rightarrow \text{random walk}$$

$$Y_t = \frac{1}{\sqrt{t}} S_t$$

$$E(Y_t) = 0 \quad V(Y_t) = \sigma^2$$

→ Proof that μ , var independent of time is not enough to say stationary or not

$$\text{cov}(Y_t, Y_{t+h})$$

$$= \text{cov}\left(\frac{1}{\sqrt{t}} \sum_{i=1}^t X_i, \frac{1}{\sqrt{t+h}} \sum_{i=1}^{t+h} X_i\right)$$

$$= \text{cov}\left(\frac{1}{\sqrt{t}} \sum_{i=1}^t X_i, \frac{1}{\sqrt{t+h}} \sum_{i=1}^t X_i + \frac{1}{\sqrt{t+h}} \sum_{i=t+1}^{t+h} X_i\right)$$

$$= \text{cov}\left(\frac{1}{\sqrt{t}} S_t, \frac{1}{\sqrt{t+h}} S_t\right) + 0 \text{ (iid)}$$

$$= \frac{t \sigma^2}{\sqrt{t} \sqrt{t+h}}$$

$$= \sigma^2 \sqrt{\frac{t}{t+h}}$$

→ dependent on both t and h .

→ 0 as $h \rightarrow \infty$.

Properties → Refer to slides

* WEAK STATIONARITY

① $\mu_t = E(X_t)$ is free from t

② $\gamma_{X_t}(X_t, X_{t+h}) = \gamma_X(h)$ if free from t , but dependent on lag h .

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$$(3) \gamma_x(h) = \gamma_x(-h)$$

- $MAC(1)$ = Moving average process of order 1

$$\{W_t\} \sim WNC(0, \sigma^2)$$

$$X_t = W_t + \theta W_{t-1}$$

$$E(X_t) = 0$$

$$\begin{aligned} \text{var}(X_t) &= \text{var}(W_t + \theta W_{t-1}) \\ &= \sigma^2 + \theta^2 \sigma^2 + 0 = (1 + \theta^2) \sigma^2 \end{aligned}$$

$$\begin{aligned} \gamma_x(1) &= \text{cov}(X_t, X_{t+1}) \\ &= \text{cov}(W_t + \theta W_{t-1}, W_{t+1} + \theta W_t) \\ &= 0 + \theta \sigma^2 + 0 + 0 = \theta \sigma^2 \end{aligned}$$

Hence

$$E(X_t) = 0$$

$$\gamma(0) = (1 + \theta^2) \sigma^2$$

$$\gamma(1) = \theta \sigma^2$$

$$\gamma(h) = 0 \quad \forall |h| > 1$$

- $AR(1)$: Autoregressive process of order 1

$$\{W_t\} \sim WNC(0, \sigma^2)$$

Assume (i) $|\phi| < 1$ and $\phi \neq 0$

(ii) $\{X_t\}$ is weakly stationary

$$X_t = \phi X_{t-1} + W_t$$

$$\begin{aligned} E(X_t) &= \phi E(X_{t-1}) + E(W_t) \\ &= \phi E(X_{t-1}) \end{aligned}$$

$$\text{var}(X_t) = \text{var}(\phi X_{t-1} + W_t)$$

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$$= \phi^2 \text{var}(X_{t-1}) + \text{var}(W_t) + 2\phi \text{cov}(X_{t-1}, W_t)$$

$$= \phi^2 \text{var}(X_{t-1}) + \sigma^2$$

$$\Rightarrow \boxed{\text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2}}$$

* Auto-correlation Coefficient

$$\rho_x(h) = \frac{\gamma_x(h)}{\gamma_x(0)} = \frac{\text{cov}(X_t, X_{t+h})}{\sqrt{\text{var}(X_t) \text{var}(X_{t+h})}}$$

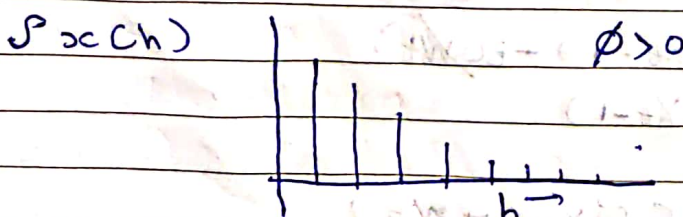
Hence for AR(1)

$$\begin{aligned} \gamma_x(h) &= \text{cov}(X_{t+h}, X_t) \quad h > 0 \\ &= \text{cov}(W_{t+h} + \phi X_{t+h-1}, X_t) \\ &= \phi \text{cov}(X_{t+h-1}, X_t) \\ &= \phi \text{cov}(W_{t+h-1} + \phi X_{t+h-2}, X_t) \\ &= \phi^2 \text{cov}(X_{t+h-2}, X_t) \\ &\vdots \\ &= \phi^h \text{cov}(X_t, X_t) = \phi^h \gamma_x(0) \end{aligned}$$

In general $\gamma_x(h) = \phi^{|h|} \gamma_x(0) = \boxed{\frac{\phi^{|h|} \sigma^2}{1 - \phi^2}}$

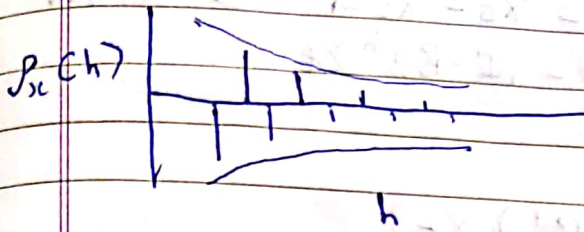
Hence

$$\rho_x(h) = \underline{\phi^{|h|}} \rightarrow \text{exponential decay.}$$



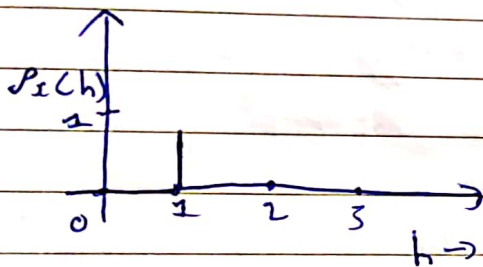
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$$-1 < \theta < 0$$



- For MAC(1) process

$$\rho_{sc}(h) = \frac{\gamma_{sc}(h)}{\gamma_{sc}(0)} = \begin{cases} 1 & h=0 \\ \theta/1-\theta & |h|=1 \\ 0 & |h| > 1 \end{cases}$$



Stops after lag 1.

$$\Gamma = \begin{matrix} & \begin{matrix} X_1 & X_2 & \dots & X_n \end{matrix} \\ \begin{matrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{matrix} & \begin{bmatrix} \gamma(0) & \gamma(1) & \gamma(2) & \dots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \gamma(1) & \dots & \gamma(n-2) \\ \gamma(2) & \gamma(1) & \gamma(0) & \dots & \gamma(n-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \gamma(n-3) & \dots & \gamma(0) \end{bmatrix} \end{matrix}$$

↓
Toeplitz matrix

* OPERATORS

• Back Shift Operator $B \rightarrow$

$$B^h X_t = X_{t-h}$$

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- Difference Operator : $I - B = \nabla$
 $\nabla X_t = X_t - X_{t-1} = (I - B) X_t$
 $\Rightarrow \nabla^h X_t = (I - B)^h X_t \rightarrow \text{Imp}$

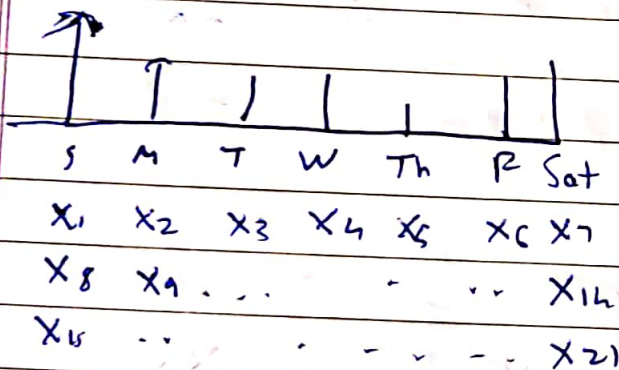
- Seasonal Difference

$$\nabla_s X_t = (I - B^s) X_t$$

$$S_t = \sum_{i=1}^t W_i \quad W_i \sim WN$$

$$\nabla S_t = S_t - S_{t-1} = W_t$$

$$\nabla S_{t-1} = S_{t-1} - S_{t-2} = W_{t-1}$$



only for sundays

$$\left. \begin{aligned} X_{25} - X_8 \\ X_8 - X_1 \end{aligned} \right\} \text{Seasonal differences}$$

* LINEAR PROCESS

$$X_t = \mu + \sum_{j=-\infty}^{+\infty} \psi_j W_{t-j}$$

where $\mu \in \mathbb{R}$, $\sum_{j=-\infty}^{+\infty} |\psi_j| < \infty$ and $W_j \sim WN(0, \sigma_w^2)$

$$E(X_t) = \mu$$

$$Y_{AC}(h) = \sigma_w^2 \sum_{j=-\infty}^{+\infty} \psi_j \psi_{j-h} < \infty$$

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$$\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$$

$$\begin{aligned} X_t &= \mu + \sum_{j=-\infty}^{\infty} \psi_j W_{t-j} \\ &= \mu + \sum_{j=-\infty}^{\infty} \psi_j (B^j W_t) \\ &= \mu + \left(\sum_{j=-\infty}^{\infty} \psi_j B^j \right) W_t \end{aligned}$$

$$= \mu + \Psi_{\infty} (B) W_t$$

~~Imp~~

To show $E(X_t) = \mu$, first check
 $E|X_t| < \infty$

$$E|X_t| = E \left| \mu + \sum_{j=-\infty}^{\infty} \psi_j W_{t-j} \right|$$

$$|x+y| \leq |x| + |y|$$

$$\leq E|\mu| + E \left| \sum_{j=-\infty}^{\infty} \psi_j W_{t-j} \right| \quad \text{Triangle inequality}$$

$$\leq |\mu| + \sum_{j=-\infty}^{\infty} |\psi_j| E|W_{t-j}|$$

$$\leq |\mu| + M \sum_{j=-\infty}^{\infty} |\psi_j| < \infty$$

$$\begin{aligned} \text{var}(W_t) &< \infty \\ \Rightarrow E(W_t^2) &< \infty \end{aligned}$$

$$\max_t E|W_t|$$

$$\text{HW) If } E(|X|^g) < \infty \text{ then } E(|X|^g) < \infty \text{ if } g \leq g$$

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then

$$\begin{aligned} E(X_t) &= E\left(\mu + \sum_{j=-\infty}^{\infty} \psi_j(B) \cdot w_t\right) \\ &= \mu + 0 \\ &= \boxed{\mu} \end{aligned}$$

To show $\gamma_x(h) = \sigma_w^2 \sum_{j=-\infty}^{\infty} \psi_j \psi_{j-h}$

First check $E|X_t X_{t+h}| < \infty$

$$\begin{aligned} &= E \left| \sum_{j=-\infty}^{\infty} \psi_j w_{t+h-j} \sum_{k=-\infty}^{\infty} \psi_k w_{t-k} \right| \\ &= E \left| \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k w_{t+h-j} w_{t-k} \right| \\ &\leq \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} |\psi_j| |\psi_k| E(|w_{t+h-j}| |w_{t-k}|) \\ &\leq m \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} |\psi_j| |\psi_k| = m \left(\sum_{j=-\infty}^{\infty} |\psi_j| \right)^2 < \infty \end{aligned}$$

Now

$$\begin{aligned} \gamma_x(h) &= \text{cov} \left(\sum_{j=-\infty}^{\infty} \psi_j w_{t+h-j}, \sum_{k=-\infty}^{\infty} \psi_k w_{t-k} \right) \\ &= \sum_j \sum_k \psi_j \psi_k \text{cov}(w_{t+h-j}, w_{t-k}) \end{aligned}$$

For a non zero contribution, it can only happen if $w_{t+h-j} = w_{t-k}$

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$$t+h-j = t-k \\ \Rightarrow k = j-h$$

$$= \sum_{j=-\infty}^{\infty} \Psi_j \Psi_{j-h} \sigma_w^2$$

* Theorem

Consider a (weakly) stationary time series $\{X_t\}$ with mean 0, and define

$$Y_t = \mu + \sum_{j=-\infty}^{\infty} \Psi_j X_{t-j}$$

where $\mu \in \mathbb{R}$ and $\sum_{j=-\infty}^{\infty} |\Psi_j| < \infty$ then
 $E(Y_t) = \mu$

Now

$$\gamma_Y(h) = \text{cov}(Y_{t+h}, Y_t)$$

$$= \text{cov}\left(\sum_{j=-\infty}^{\infty} \Psi_j X_{t+h-j}, \sum_{k=-\infty}^{\infty} \Psi_k X_{t-k}\right)$$

$$= \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \Psi_j \Psi_k E(X_{t+h-j}, X_{t-k})$$

$$= \sum_j \sum_k \Psi_j \Psi_k \text{cov}(X_{t+h-j}, X_{t-k})$$

$$= \sum_j \sum_k \Psi_j \Psi_k \gamma_X(h+k-j)$$

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- For MAC(1) - To write w as a linear process

$$X_t = W_t + \theta W_{t-1}$$

$\{W_t\}$ is a W.N. sequence

$$\begin{aligned} \psi_j &= 0 \text{ if } j < 0 \\ \psi_0 &= 1 \\ \psi_1 &= \theta \\ \psi_2 &= 0 \\ \psi_3 &= 0 \\ &\vdots \end{aligned}$$

- To express AR(1) process as a linear process

$$\begin{aligned} \psi_j &= 0 \text{ if } j < 0 \\ \psi_0 &= 1 \\ \psi_1 &= \phi \\ \psi_2 &= \phi^2 \\ &\vdots \\ \psi_j &= \phi^j \end{aligned} \quad \mu = 0$$

- Random Walk

$\{W_t\} \sim (N(0, \sigma^2))$

$$\begin{aligned} \psi_0 &= 0 \\ \psi_1 &= 1 \\ \psi_2 &= 1 \\ &\vdots \end{aligned} \quad S_t = \sum_{i=1}^t W_i$$

* PROPERTIES OF ACF

$$\rho(h) = \frac{\gamma_x(h)}{\gamma_x(0)}$$

• We have $\rho(h) = \rho(-h)$

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Let $1 \leq i, j \leq n$ and $R = ((\rho(|i-j|)))_{i,j}$ then

$$a^T R a \geq 0 \quad \forall a \in \mathbb{R}^n$$

$R \rightarrow$ p.s.d and $\delta(\cdot)$ and $\rho(\cdot)$ are positive semidefinite fⁿ.

* MOVING AVERAGE PROCESSES

- $MAC(q)$ is a moving avg process of order q is defined as

$$X_t = W_t + \theta_1 W_{t-1} + \dots + \theta_q W_{t-q}$$

Linear process with $\psi_0 = 1, \psi_j = \theta_j$

* AUTO - REGRESSIVE PROCESS

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + W_t$$

- We can consider an AR(1) as an MAC(∞) process.

$$\lim_{k \rightarrow \infty} E \left(X_t - \sum_{j=0}^k \phi^j W_{t-j} \right)^2 = 0$$

$$E \left(X_t - \sum_{j=0}^{\infty} \phi^j W_{t-j} \right)^2 \text{ (Covariance)}$$

$$\begin{aligned} X_t &= \phi X_{t-1} + W_t \\ &= \phi (\phi W_{t-1} + \phi X_{t-2}) + W_t \end{aligned}$$

$$= W_t + \phi W_{t-1} + \phi^2 X_{t-2}$$

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$$= \underbrace{\left(\sum_{j=1}^k \phi^j w_{t-j} \right)}_{\text{MACH}} + \phi^{k+1} x_t - (1-\phi) x_t$$

$$\lim_{k \rightarrow \infty} E \left(x_t - \sum_{j=0}^k \phi^j w_{t-j} \right)^2$$

$$= \lim_{k \rightarrow \infty} \left(\phi^{k+1} \cdot x_t - (1-\phi) x_t \right)^2$$

$$\lim_{k \rightarrow \infty} \phi^{2k+1} E(x_t^2 - 2\phi x_t + \phi^2)$$

$L_2 \rightarrow$ Mean square convergence

$L_1 \rightarrow$ $\rho(x_t)$

distribution convergence $1/\phi^{k+1}$



→
 - Use of Sample ACF:-

Time Series	ACF
White noise	Zero
MA(1)	zero for $ h > 1$
AR(1)	Decays to zero exponentially

* AUTOREGRESSIVE MOVING AVERAGE PROCESS

$$\text{MA(1)} \Rightarrow X_t = W_t + \theta W_{t-1} = (I + \theta B) W_t \\ = \Phi_1(B) W_t$$

$$\text{AR(1)} \Rightarrow X_t = \phi X_{t-1} + W_t \\ \Rightarrow W_t = X_t - \phi X_{t-1} = (I - \phi B) X_t \\ = \Phi_1(B) X_t$$

$\Rightarrow \text{ARMA(1,1)}$ is defined as: $\Phi_1(B) X_t = \Theta_1(B) W_t$
 $\Rightarrow X_t = \frac{\Theta_1(B)}{\Phi_1(B)} W_t$
 $\hookrightarrow$ Ratio of 2 polynomials.

$$X_t = \left(\frac{I + \theta B}{I - \phi B} \right) W_t = (I + \theta B) (I - \phi B)^{-1} W_t \\ = (I + \theta B) \left(\sum_{j=0}^{\infty} \phi^j B^j \right) W_t \\ \Rightarrow \Psi(B) = \left(\sum_{j=0}^{\infty} \psi_j B^j \right) W_t$$

where $X_t = \Psi(B) W_t$
 $\psi_0 = 1 \quad \psi_j = (\theta + \phi) \phi^{j-1} \quad j \geq 1$

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If $\phi + \theta = 0$, then X_t is white noise
 $\theta = -\phi \Leftrightarrow \frac{\phi}{\phi} = \frac{-\phi}{-\phi}$

Now

$$\phi_p(z) = 1 - \sum_{i=1}^p \phi_i z^i \quad \text{if } \phi_p \neq 0$$

$$\theta_q(z) = 1 - \sum_{i=1}^q \theta_i z^i \quad \text{if } \theta_q \neq 0$$

Then, ARMA(p, q) model can be written as

$$\phi_p(B) X_t = \theta_q(B) W_t$$

$$1) Y_x(0) = \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 = ?$$

$$2) Y_x(1) = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+1} = ?$$

$$3) Y_x(h) = [\quad] Y_x(1) \quad h \geq 1$$

$$\textcircled{1} \quad \sigma^2 \left[1 + (\theta + \phi)^2 \sum_{j=1}^{\infty} \phi^{2(j-1)} \right] \quad \hookrightarrow 1 + \phi^2 + \phi^4 + \dots = \frac{1}{1 - \phi^2}$$

$$= \sigma^2 \left[1 + \frac{(\theta + \phi)^2}{1 - \phi^2} \right]$$

$$\textcircled{2} \quad \sigma^2 \left[\psi_1 + \sum_{j=1}^{\infty} (\theta + \phi)^2 \phi^{j-1} \phi \right]$$

$$\sigma^2 \left[(\theta + \phi) + \frac{(\theta + \phi)^2 \phi}{1 - \phi^2} \right]$$

$$\textcircled{3} \quad Y_x(h) = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h} = \phi^{h-1} Y_x(1)$$

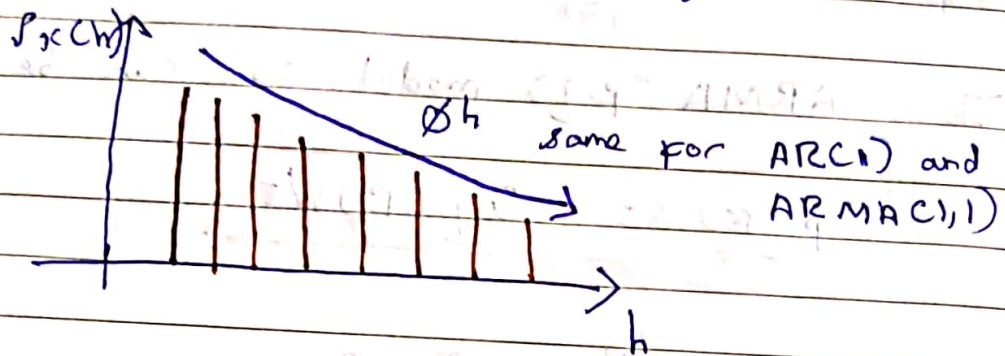
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Autocorrelation coeff:-

• $\rho_{xx}(0) = 1$

•
$$\rho_{xx}(h) = \frac{\gamma_{xx}(h)}{\gamma_{xx}(0)} = \frac{\phi^{h-1} \gamma_{xx}(1)}{\gamma_{xx}(0)}$$

$$= \phi^h \left(\frac{\gamma_{xx}(1)}{\gamma_{xx}(0)} \right)$$



Estimation for weakly stationary Time Series:-

- Let $\{X_t\}$ be a weakly stationary time series

• $E(X_t) = \mu$

$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$

$E(\hat{\mu}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \mu \rightarrow$ unbiased estimate

• $V(\bar{X}) = ? = E[(\bar{X} - \mu)^2] - E(E(\bar{X} - \mu))^2$

$\downarrow$

$= E[(\bar{X} - \mu)^2]$

$\hookrightarrow \bar{X} - \mu = \frac{1}{n} \sum (X_i - \mu)$

$= \frac{1}{n} \mathbf{1}^T (\mathbf{X} - \mu \mathbf{1})$

$= \text{cov}\left(\frac{1}{n} \mathbf{1}^T (\mathbf{X} - \mu \mathbf{1}), \frac{1}{n} \mathbf{1}^T (\mathbf{X} - \mu \mathbf{1})\right)$

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$$= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{cov}(X_i, X_j)$$

$$= \frac{1}{n^2} \mathbf{1}^T \mathbf{\Gamma}_X \mathbf{1}$$

$$\cdot \text{DC}(X) = \mathbf{\Gamma}_X = \begin{bmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \dots & \gamma(0) \end{bmatrix}$$

$$= \left(\left(\gamma_{|i-j|} \right) \right)_{ij}$$

$$= \frac{1}{n^2} \left[n \cdot \gamma(0) + 2(n-1)\gamma(1) + 2(n-2)\gamma(2) + \dots + 2\gamma(n-1) \right]$$

$$= \frac{1}{n} \sum_{h=-(n-1)}^{n-1} \left(1 - \frac{|h|}{n} \right) \gamma(h)$$

$$= \frac{1}{n} \sum_{h=-n}^n \left(1 - \frac{|h|}{n} \right) \gamma(h)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \text{var}(\bar{X}) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{h=-n}^n \left(1 - \frac{|h|}{n} \right) \gamma_e(h)$$

$$\leq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{h=-n}^n \left(1 - \frac{|h|}{n} \right) |\gamma_e(h)|$$

$$\leq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{h=-n}^n |\gamma_e(h)|$$

$$\text{If } \sum_{h=-\infty}^{\infty} |\gamma_e(h)| < \infty, \text{ then } \lim_{n \rightarrow \infty} \text{var}(\bar{X}) \downarrow 0$$

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$$\sqrt{n}(\bar{X} - \mu) \sim N\left(0, \sum_{|h| \leq n} \left(1 - \frac{|h|}{n}\right) \gamma(h)\right)$$

as $n \rightarrow \infty$

In practice,

$$\sqrt{n}(\bar{X} - \mu) \sim N\left(0, \sum_{|h| \leq [\sqrt{n}]} \left(1 - \frac{|h|}{n}\right) \gamma(h)\right)$$

95% CI of μ :

$$\left(\bar{X} - 1.96 \frac{\hat{\sigma}}{\sqrt{n}}, \bar{X} + 1.96 \frac{\hat{\sigma}}{\sqrt{n}} \right)$$

where $\hat{\sigma}^2 = \sum_{|h| \leq \sqrt{n}} \left(1 - \frac{|h|}{n}\right) \gamma(h)$

Define

$$\tilde{X}_t = X_t - \mu = X_t - \bar{X}$$

$$M = \begin{pmatrix} 0 & 0 & \dots & 0 & \tilde{X}_1 & \tilde{X}_2 & \dots & \tilde{X}_n \\ 0 & 0 & \dots & 0 & \tilde{X}_1 & \tilde{X}_2 & \dots & \tilde{X}_n \\ \vdots & \vdots & \dots & \vdots & \tilde{X}_1 & \tilde{X}_2 & \dots & \tilde{X}_n \\ \tilde{X}_1 & \tilde{X}_2 & \dots & \tilde{X}_n & 0 & 0 & \dots & 0 \end{pmatrix}$$

$n \times 2n-1$

$$\Gamma = \left(\gamma(|i-j|) \right)_{i,j} \Rightarrow \hat{\Gamma} = \frac{1}{n} M M^T$$

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{i=1}^{n-|h|} (X_{i+|h|} - \bar{X})(X_i - \bar{X})$$

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HW

- $\hat{F}$ is symmetric
- $\hat{F}$ is psd

- Estimated Auto correlation coeff:-

$$\hat{\rho}_x(h) = \frac{\hat{\gamma}_x(h)}{\hat{\gamma}_x(0)}$$

- For a linear process if $\sum_{-\infty}^{\infty} |\psi_i| < \infty$ and $\sum_0^{\infty} h |\psi_h|^2 < \infty$

Then we have limiting distribution of

$$\hat{\rho} = (\hat{\rho}(1), \hat{\rho}(2), \dots, \hat{\rho}(h))$$

- Large sample Distribution:-

$$\sqrt{n} (\hat{\rho} - \rho) \sim N(0, W)$$

$$W_{ij} = \text{cov}(\sqrt{n}(\hat{\rho}_i - \rho_i), \sqrt{n}(\hat{\rho}_j - \rho_j))$$

$$= \sum_{k=1}^{\infty} \left[\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k) \right] \times$$

$$\left[\rho(k+j) + \rho(k-j) - 2\rho(k)\rho(j) \right]$$

↳ **Bartlett's Formula**

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PARTIAL CORRELATION COEFF

$$\begin{pmatrix} Y \\ Z \\ \underline{X} \\ \sim \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_Y \\ \mu_Z \\ \underline{\mu_X} \\ \sim \end{pmatrix}, \begin{pmatrix} \Sigma_{YY} & \Sigma_{YZ} & \Sigma_{YX} \\ \Sigma_{YZ} & \Sigma_{ZZ} & \Sigma_{ZX} \\ \Sigma_{YX} & \Sigma_{ZX} & \Sigma_{XX} \end{pmatrix} \right)$$

$$E(Y) = \mu_Y$$

$$E(Z) = \mu_Z$$

$$E(\underline{X}) = \underline{\mu_X}$$

$$D \begin{pmatrix} Y \\ Z \end{pmatrix} = \Sigma_{YZ} = \begin{pmatrix} \sigma_{YY} & \sigma_{YZ} \\ \sigma_{YZ} & \sigma_{ZZ} \end{pmatrix}$$

$$D(\underline{X}) = (\sigma_{X_i X_j})$$

Define

$$\sigma_{YX} = \text{cov}(Y, \underline{X}) = \begin{pmatrix} \sigma_{YX_1} \\ \sigma_{YX_2} \\ \vdots \\ \sigma_{YX_p} \end{pmatrix}$$

$$\sigma_{ZX} = \text{cov}(Z, \underline{X}) = \begin{pmatrix} \sigma_{ZX_1} \\ \sigma_{ZX_2} \\ \vdots \\ \sigma_{ZX_p} \end{pmatrix}$$

$$E(Y|X=x)$$

$$\Sigma_{YZX} = \begin{pmatrix} \sigma_{YX_1} & \sigma_{ZX_1} \\ \sigma_{YX_2} & \sigma_{ZX_2} \\ \vdots & \vdots \\ \sigma_{YX_p} & \sigma_{ZX_p} \end{pmatrix} = (\sigma_{YX} \quad \sigma_{ZX})_{p \times 2}$$

$$\sigma_{YZ.X} = \text{cov}(Y - E(Y|X), Z - E(Z|X))$$

↓

$$\sqrt{V(Y - E(Y|X)) V(Z - E(Z|X))}$$

Partial correlation coeff. between (Y, Z) given X

↓

correlation between residuals after removing the effect of X on Y and Z.

$E(Y|X)$ is the regression of

Y on X and $E(Z|X)$ is the

regression of Z on X.

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- Conditional distribution of $\begin{pmatrix} Y \\ Z \end{pmatrix}$ given $X = x$ is

$$N \left(\begin{pmatrix} \mu_y \\ \mu_z \end{pmatrix} + \sum_{YZX}^T \Sigma_X (x - \mu_x), \Sigma_{YZ} - \sum_{YZX}^T \Sigma_X^{-1} \sum_{YZX} \right)$$

Regression of $\begin{pmatrix} Y \\ Z \end{pmatrix}$ vector given $X = x$ vector

$$\begin{aligned} (X, Y) &\sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho) \\ \Rightarrow Y|X &\sim N \left(\mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x), (1 - \rho^2) \sigma_y^2 \right) \end{aligned}$$

Bivariate structure special case:

$$\mu_y + \frac{\rho \sigma_y \sigma_x}{\sigma_x^2} (x - \mu_x) = \mu_y + \frac{\text{cov}(X, Y)}{\text{var}(X)} (x - \mu_x)$$

$$\begin{aligned} \begin{pmatrix} \sigma_y^2 & \rho \sigma_y \sigma_x \\ \rho \sigma_x \sigma_y & \sigma_x^2 \end{pmatrix} &= \begin{pmatrix} \sigma_y^2 & \rho \sigma_y \sigma_x \\ \rho \sigma_x \sigma_y & \sigma_x^2 \end{pmatrix} = \sigma_y^2 \begin{pmatrix} 1 & \rho \frac{\sigma_x}{\sigma_y} \\ \rho \frac{\sigma_y}{\sigma_x} & \frac{\sigma_x^2}{\sigma_y^2} \end{pmatrix} \\ &= \sigma_y^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = (1 - \rho^2) \sigma_y^2 \end{aligned}$$

here $Y \sim Z$

Errors: $Z \sim N(\mu, \sigma^2)$
 $\Rightarrow Z - \mu \sim N(0, \sigma^2)$

$$\left[\begin{pmatrix} Y \\ Z \end{pmatrix} | X \right] = E \left[\begin{pmatrix} Y \\ Z \end{pmatrix} | X = x \right] \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma_{YZ} - \sum_{YZX}^T \Sigma_X^{-1} \sum_{YZX} \right)$$

$$\text{Cov}(\rho_{Y|X}, \rho_{Z|X}) = \sigma_{YZ} - \sigma_{YX}^T \Sigma_X^{-1} \sigma_{ZX}$$

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$$\Sigma_{YZ} = \begin{pmatrix} \sigma_{YY} & \sigma_{YZ} \\ \sigma_{YZ} & \sigma_{ZZ} \end{pmatrix}$$

$$\sigma_{YX} = \text{cov}(Y, \underline{X}) = \begin{pmatrix} \sigma_{YX_1} \\ \sigma_{YX_2} \\ \vdots \\ \sigma_{YX_K} \end{pmatrix}$$

$$\sigma_{ZX} = \text{cov}(Z, \underline{X}) = \begin{pmatrix} \sigma_{ZX_1} \\ \vdots \\ \sigma_{ZX_K} \end{pmatrix}$$

$$\Sigma_{YZ|X} = \text{cov}(Y|X, Z|X)$$

Partial correlation coeff (Y, Z) given X:

$$r_{YZ|X} = \frac{\text{cov}(e_{Y|X}, e_{Z|X})}{\sqrt{v(e_{Y|X}) v(e_{Z|X})}}$$

- If $\begin{pmatrix} Y \\ Z \\ \underline{X} \end{pmatrix}$ does not follow normal distribution,

then instead of conditional distribution of $\begin{pmatrix} Y \\ Z \end{pmatrix}$ given $\underline{X} = \underline{x}$ we should find best linear predictor / estimator.

$$E \begin{pmatrix} Y \\ Z \\ \underline{X} \end{pmatrix} = \begin{pmatrix} \mu_Y \\ \mu_Z \\ \underline{\mu_X} \end{pmatrix} \quad D \begin{pmatrix} Y \\ Z \\ \underline{X} \end{pmatrix} = \begin{pmatrix} \Sigma_{YZ} & \Sigma_{YZX}^T \\ \Sigma_{YZX} & \Sigma_X \end{pmatrix}$$

- If we consider the Best Linear predictor of $\begin{pmatrix} Y \\ Z \end{pmatrix}$ given $\underline{X}$ and compute the error, then the structure of partial correl. coeff will remain unchanged. Only the normality will not be there.

$$\begin{pmatrix} Y \\ Z \\ \underline{X} \end{pmatrix} \Rightarrow \begin{pmatrix} \mu_Y \\ \mu_Z \\ \underline{\mu_X} \end{pmatrix} \begin{pmatrix} \sigma_{XX} & \sigma_{XX}^T & \sigma_{YZ} \\ \sigma_{YX} & \Sigma_X & \sigma_{ZX} \\ \sigma_{YZ} & \sigma_{ZX}^T & \sigma_{ZZ} \end{pmatrix} \begin{matrix} \text{--- This exp} \\ \text{is useful} \\ \text{for time} \\ \text{series} \end{matrix}$$

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PACF:-

Let $\{x_t\}$ be a weakly stationary time series
Consider $(x_0, x_1, \dots, x_h, x_{h+1})$

Correlation between the error in linear prediction of x_0 on $(x_1, \dots, x_h)$ and the error in linear prediction of x_{h+1} on $(x_1, \dots, x_h)$ is known as the PACF of order $h+1$ denoted by $\alpha(h+1)$

$$\begin{array}{c|cccc} & x_0 & x_1 & \dots & x_h & x_{h+1} \\ \hline x_0 & \gamma(0) & \gamma(1) & \dots & \gamma(h) & \gamma(h+1) \\ x_1 & \gamma(1) & \gamma(0) & \dots & \gamma(h-1) & \gamma(h) \\ \vdots & \gamma(2) & \vdots & \ddots & \vdots & \gamma(h-1) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_h & \gamma(h) & \gamma(h-1) & \dots & \gamma(0) & \gamma(1) \\ x_{h+1} & \gamma(h+1) & \gamma(h) & \dots & \gamma(1) & \gamma(0) \end{array} \quad - (2)$$

comparing (2) and (2)

Partial correl coefft of (y, z) given $\tilde{x}$

$$\rho_{yz|x} = \frac{\text{cov}(e_{y|x}, e_{z|x})}{\sqrt{v(e_{y|x}) v(e_{z|x})}}$$

$\rightarrow$ scalar

$$= \frac{\gamma(h+1) - \tilde{\gamma}_h^T(c_1) \Gamma_h^{-1} \gamma_h(c_1)}{\gamma(0) - \tilde{\gamma}_h^T(c_1) \Gamma_h^{-1} \tilde{\gamma}_h(c_1)}$$

where

$$\gamma_h(c_1) = (\gamma(c_1), \gamma(c_2), \dots, \gamma(c_h))$$

$$\tilde{\gamma}_h(c_1) = (\gamma(c_h), \gamma(c_{h-1}), \dots, \gamma(c_1))$$

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Note : $\text{var}(e_y | \tilde{x}) = \sigma_{yy} - \sigma_{y\tilde{x}}^T \Sigma_{\tilde{x}}^{-1} \sigma_{y\tilde{x}}$

$\text{var}(e_z | \tilde{x}) = \sigma_{zz} - \sigma_{z\tilde{x}}^T \Sigma_{\tilde{x}}^{-1} \sigma_{z\tilde{x}}$

$\tilde{y}_h(c) \Gamma_h^{-1} \tilde{y}_h(c) = \tilde{y}_h(c) \Gamma_h^{-1} \tilde{y}_h(c)$

$\tilde{y}_h(c) = (\gamma(c), \gamma(c+1), \dots, \gamma(c+h))^T$

$\tilde{y}_h(c) = (\gamma(c), \gamma(c+1), \dots, \gamma(c+h))^T$

$\tilde{y}_h(c) = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 0 & \dots & 0 & 0 \end{bmatrix} \begin{pmatrix} \gamma(c) \\ \gamma(c+1) \\ \vdots \\ \gamma(c+h) \end{pmatrix} = \begin{bmatrix} \gamma(c) \\ \gamma(c+1) \\ \vdots \\ \gamma(c+h) \end{bmatrix}$

$\downarrow$
 $R_h(c)$

$\tilde{y}_h(c) = R_h \tilde{y}_h(c)$

$\begin{pmatrix} \alpha_{c+h+1} \end{pmatrix} = \frac{\gamma(c+h) - \tilde{y}_h(c) \Gamma_h^{-1} \tilde{y}_h(c)}{\gamma(c) - \tilde{y}_h(c) \Gamma_h^{-1} \tilde{y}_h(c)}$

$\rightarrow$ PACF of $\log C_{h+1}$

$|\alpha_{c+h+1}| \leq 1$

$\tilde{y}_h^T(c) \Gamma_h^{-1} \tilde{y}_h(c) = \gamma^T(c) [R_h^T \Gamma_h^{-1} R_h] \gamma(c)$

$= \gamma^T(c) \Gamma_h^{-1} \gamma(c)$

We can classify between ARMA, MA, AR through ACF and PACF together.

Forecasting Weakly Stationary Time Series:

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Assumptions:

- ① $\{X_t\}$ is a weakly stationary T.S.
 - ② 'n' observations available
 - ③ $E(X_t) = \mu$ known
 - ④ $\Gamma = (| \gamma_{C(i-j)} |)_{i,j}$ known
- } We will relax these assumptions later

Goal: Given $\{x_1, x_2, \dots, x_n\}$ we want to predict X_{n+h} , $h \geq 1$ such that the least square error is minimised for linear predictor

$$\hat{X}_{n+h} = a^T \tilde{x}$$

MLR:-

$$Y \Rightarrow \hat{X}_{n+h}$$

Assume

$$E(Y) = 0$$

$$E(X) = 0$$

$$V(Y) = \sigma_{yy}$$

$$\text{cov}(X, Y) = \sigma_{yx}$$

$$D(X) = \sum x \text{ are known}$$

Best linear predictor for Y given $X = \tilde{x}$ is $\beta^T \tilde{x}$ which minimises the least square condition.

$$s^2 = E(Y - \beta^T \tilde{x})^2 \quad \text{Set } \frac{\partial s^2}{\partial \beta} \bigg|_{\hat{\beta}} = 0$$

$$E(Y \tilde{x}) = E(X \tilde{x}^T) \hat{\beta}$$

$$\sigma_{yx} = \sum \tilde{x} \hat{\beta} \quad (\text{Normal equation})$$

$$\hat{\beta}_{LS} = \Sigma \tilde{x}^{-1} \sigma_{yx} \rightarrow \text{Best linear predictor generally.}$$

Syde
Gyde
!

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Result:

$$\text{var}(Y - \hat{\beta}^T X) = \sigma_{YY} - \sigma_{YX}^T \Sigma_X^{-1} \sigma_{YX}$$

Proof:

$$\begin{aligned} \text{var}(Y - \hat{\beta}^T X) &= \text{var}(Y) + \text{var}(\hat{\beta}^T X) - 2\text{cov}(\cdot) \\ &= \sigma_{YY} + \hat{\beta}^T \Sigma_X \hat{\beta} - 2\hat{\beta}^T \sigma_{YX} \\ &= \sigma_{YY} + \sigma_{YX}^T \Sigma_X^{-1} \Sigma_X \Sigma_X^{-1} \sigma_{YX} - 2\sigma_{YX}^T \Sigma_X^{-1} \sigma_{YX} \end{aligned}$$

$$= \sigma_{YY} + \underbrace{\sigma_{YX}^T \Sigma_X^{-1} \sigma_{YX}} - 2\sigma_{YX}^T \Sigma_X^{-1} \sigma_{YX}$$

$$= \boxed{\sigma_{YY} - \underbrace{\sigma_{YX}^T \Sigma_X^{-1} \sigma_{YX}}}$$

- Best linear predictor is unique

proof:- Assume that there are more than one best linear predictor $\hat{\beta}$ and $\hat{\beta}_0$

$$\begin{aligned} \text{var}(\hat{\beta}^T X - \hat{\beta}_0^T X) &= \text{var}((\hat{\beta} - \hat{\beta}_0)^T X) \\ &= (\hat{\beta} - \hat{\beta}_0)^T \Sigma_X (\hat{\beta} - \hat{\beta}_0) \end{aligned}$$

$\downarrow$
 $= 0$ Both $\hat{\beta}$ and $\hat{\beta}_0$ should satisfy normal equations

$$E((\hat{\beta} - \hat{\beta}_0)^T X) = 0$$

$$\left. \begin{aligned} \sigma_{YX} &= \Sigma_X \hat{\beta} \\ \sigma_{YX} &= \Sigma_X \hat{\beta}_0 \end{aligned} \right\} \Sigma_X (\hat{\beta} - \hat{\beta}_0) = 0$$

$$\text{Then } E(\hat{\beta}^T X - \hat{\beta}_0^T X) = 0$$

$$- [E((Y - \hat{\beta}^T X)^2)] = \frac{\text{var}(Y - \hat{\beta}^T X)}{\text{var}(Y)}$$

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proof:-

$$[\text{corr}(Y, \hat{\beta}^T X)]^2 = \frac{(\text{cov}(Y, \hat{\beta}^T X))^2}{\text{var}(Y) \text{var}(\hat{\beta}^T X)}$$

$$= \frac{(\hat{\beta}^T \sigma_{YX})^2}{\sigma_{YY} (\hat{\beta}^T \Sigma_X \hat{\beta})}$$

$$= \boxed{\frac{\sigma_{YX}^T \Sigma_X^{-1} \sigma_{YX}}{\sigma_{YY}}}$$

- Show that $\text{cov}(Y - \hat{\beta}^T X, X_j) = 0$ $1 \leq j \leq n$

proof: $e_j = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ $X_j = e_j^T X$

$$\text{cov}(Y - \hat{\beta}^T X, e_j^T X)$$

$$= \text{cov}(Y, e_j^T X) - \text{cov}(\hat{\beta}^T X, e_j^T X)$$

$$= e_j^T \sigma_{YX} - \hat{\beta}^T \Sigma_X e_j$$

$$= e_j^T \sigma_{YX} - \sigma_{YX}^T e_j$$

$$= \boxed{0}$$

Consider $\{X_t\}$ be a weakly stationary time series
 $E(X_t) = \mu$. We want to predict $Y = X_{n+h}$
 based on $(X_n, X_{n-1}, \dots, X_1) = \tilde{X}$

$$E(X) = \begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{pmatrix}$$

$$E(X_{n+h}) = \mu$$

$$D \begin{pmatrix} Y \\ \tilde{X} \end{pmatrix} = D \begin{pmatrix} X_{n+h} \\ X_n \\ \vdots \\ X_1 \end{pmatrix}$$

$$= \begin{pmatrix} \gamma_0 & \gamma(h) & \gamma(h+1) & \dots & \gamma(h+n-1) \\ \gamma(h) & \gamma_0 & \gamma(1) & \dots & \gamma(n-1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma(h+n-1) & \gamma(n-1) & \gamma(n-2) & \dots & \gamma_0 \end{pmatrix}$$

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$$= \begin{pmatrix} \sigma_{yy} & \sigma_{yx}^T \\ \sigma_{yx} & \Sigma_x \end{pmatrix} \quad \text{cov} \begin{pmatrix} y - \mu_y \\ x - \mu_x \end{pmatrix}$$

So, the predicted value of X_{n+h} will be

$$\begin{aligned} \hat{X}_{n+h} &= \mu + a^T (X - \mu) \\ &= \mu + \sum_{i=1}^n a_i (X_{n+1-i} - \mu) \end{aligned}$$

$$a^T = (a_1, a_2, \dots, a_n)$$

$$\tilde{X}^T = (X_n, X_{n-1}, \dots, X_1)$$

① The coefficient vector of best linear predictor is

$$\hat{a} = \Gamma_n^{-1} \gamma_n Ch \quad \text{as } \hat{\beta} = \Sigma_x^{-1} \sigma_{yx} / (\gamma(Ch), \gamma(Ch+1), \dots, \gamma(Ch+h-1)) = \tilde{\gamma}_n Ch$$

② Expected square error in prediction

$$\begin{aligned} E[(X_{n+h} - \mu) - a^T (X - \mu)]^2 \\ = \gamma(Ch) - \tilde{\gamma}_n Ch^T \Gamma_n^{-1} \tilde{\gamma}_n Ch \end{aligned}$$

$$\text{as } \sigma_{yx} = \sigma_{yy} - \sigma_{yx} \Sigma_x^{-1} \sigma_{yx}^T$$

③ Expected value of prediction error

$$E[(X_{n+h} - \mu) - a^T (X - \mu)] = 0$$

④ Correlation between predictor variables and the prediction error

$$\begin{aligned} \text{corr}[(X_{n+h} - \mu) - \hat{a}^T (X - \mu), (X_j - \mu)] &= 0 \\ \text{corr}[(X_{n+h} - \mu) - \hat{a}^T (X - \mu), b^T (X - \mu)] &= 0 \end{aligned}$$

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eg) Consider

$\{X_t\} \sim \text{AR}(1)$ $|\phi| < 1$ $\phi \neq 0$
with observations $(X_1, X_2, \dots, X_n)$ known

$$\hat{X}_{n+1} = \hat{a} \hat{x} \quad E(X) = 0$$

$$\begin{pmatrix} y(0) & \dots & y(n-1) \\ \vdots & & \vdots \\ y(n-1) & \dots & y(0) \end{pmatrix} \hat{a} = \begin{pmatrix} y(1) \\ y(2) \\ \vdots \\ y(n) \end{pmatrix}$$

$$\frac{\sigma^2}{1-\phi^2} \begin{pmatrix} 1 & \phi & \phi^2 & \dots & \phi^{n-1} \\ \vdots & & & & \vdots \\ \phi^{n-1} & \dots & \dots & \dots & 1 \end{pmatrix} \hat{a} = \begin{pmatrix} \phi \\ \phi^2 \\ \vdots \\ \phi^n \end{pmatrix} \frac{\sigma^2}{1-\phi^2}$$

$$a = \begin{pmatrix} \phi \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$b = \begin{pmatrix} \phi^n \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$X_t = \phi X_{t-1} + W_t$$

H.W.

$$X_{n+h} = \hat{b}^T \hat{x} \rightarrow \text{What is the value of } \hat{b}$$

$$E(X_{n+h} - \hat{b}^T \hat{x})^2 = ??$$

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Application :-

Missing value problem (in time series)

AR(1): $X_1, X_2, \dots, X_3$ ——— $X_5, X_6, X_7, \dots$

Several ways to find X_4

- (i) From X_1, X_2, X_3 , — try to predict X_4
- (ii) From X_7, X_6, X_5 try to predict X_4
- (iii) From X_3, X_5 try to predict X_4

$$\begin{matrix} & X_3 & X_4 & X_5 \\ \begin{matrix} X_2 \\ X_4 \\ X_5 \end{matrix} & \begin{pmatrix} \phi(0) & \gamma(1) & \gamma(2) \\ \gamma(1) & \gamma(0) & \gamma(1) \\ \gamma(2) & \gamma(1) & \gamma(0) \end{pmatrix} & \times & \begin{pmatrix} \gamma(0) & \gamma(1) \\ \gamma(1) & \gamma(0) \end{pmatrix} & \hat{a} = \begin{pmatrix} \gamma(1) \\ \gamma(0) \end{pmatrix} \end{matrix}$$

$$\frac{\sigma^2}{1-\phi^2} \begin{pmatrix} 1 & \phi^2 \\ \phi^2 & 1 \end{pmatrix} \hat{a} = \frac{\sigma^2}{1-\phi^2} \begin{pmatrix} \phi \\ \phi \end{pmatrix}$$

$$\hat{a} = \frac{1}{1+\phi^2} \begin{pmatrix} \phi \\ \phi \end{pmatrix} \begin{pmatrix} X_3 \\ X_5 \end{pmatrix}$$

$$\frac{\phi(X_3 + X_5)}{1+\phi^2} \neq \frac{X_3 + X_5}{2}$$

$$\begin{pmatrix} X_3 & X_5 \end{pmatrix} \begin{pmatrix} \phi \\ \phi \end{pmatrix} \text{ or } \hat{a}^T \begin{pmatrix} X_3 \\ X_5 \end{pmatrix}$$

There were simply taking the average to replace the missing value might not be the best option. If we know the structure, we can use the above formula

$$\frac{\phi(X_3 + X_5)}{1+\phi^2} \text{ which will be advantageous}$$

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Purkin-Levenson Algorithm

Aim :- To obtain the coefficient vector for best linear prediction in a recurrent iteration from the previous one.

For $\tilde{a}_{n-1}^{(old)} = \tilde{a}_{n-1}^{(1)}$

$\tilde{a}_{n-1}^{(old)}$ is known when we predict X_{n+1} based on $(X_n, X_{n-1}, \dots, X_2)$

Now we want to predict X_{n+1} based on $(X_n, X_{n-1}, \dots, X_1)$

We need to estimate $\tilde{a}_n = (\tilde{a}_{n-1}^{updated}, a_n^{new})$

$$\begin{bmatrix} \tilde{a}_{n-1}^{(1)} & \tilde{a}_{n-1}^{updated} \\ \left(\tilde{a}_{n-1}^{(1)} \right)^T & a_n \end{bmatrix} \begin{bmatrix} \tilde{y}_{n-1}^{(1)} \rightarrow b \\ \tilde{y}_{n-1}^{(0)} \rightarrow d \end{bmatrix} = \begin{bmatrix} \tilde{y}_{n-1}^{(1)} \\ \dots \\ \tilde{y}_{n-1}^{(n)} \end{bmatrix}$$

reverse

$$\tilde{y}_{n-1}^{(1)} = (\tilde{y}_{n-1}^{(1)} \dots \tilde{y}_{n-1}^{(n-1)})$$

$$\tilde{y}_{n-1}^{(0)} = (\tilde{y}_{n-1}^{(0)} \dots \tilde{y}_{n-1}^{(0)}) = \tilde{r}_1 \tilde{y}_{n-1}^{(1)}$$

$$\begin{bmatrix} A & b \\ c^T & d \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + \frac{A^{-1} b c^T A^{-1}}{d - c^T A^{-1} b} & -\frac{A^{-1} b}{d - c^T A^{-1} b} \\ -\frac{c^T A^{-1}}{d - c^T A^{-1} b} & \frac{1}{d - c^T A^{-1} b} \end{bmatrix}$$

block matrix inversion

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$$\hat{a}_n = \frac{y(n) - (y_{n-1}^{old})^T \Gamma_{n-1}^{-1} y_{n-1}^{old}}{y(n) - (y_{n-1}^{old})^T \Gamma_{n-1}^{-1} y_{n-1}^{old}}$$

$$= y(n) - (y_{n-1}^{old})^T \hat{a}_{n-1}^{old}$$

$$y(n) - (y_{n-1}^{old})^T \hat{a}_{n-1}^{old}$$

- * a_n is a PACF of order $(n) \Rightarrow |a_n| \leq 1$
- * a_n can be expressed with a_{n-1}^{old}

$$a_{n-1}^{updated} = a_{n-1}^{old} - a_n (a_{n-1}^{old})^T$$

Error analysis

$$E \left(X_{n+1} - (a_{n-1}^{old})^T X_{n:1} \right)^2$$

$$= y(n) - (y_{n-1}^{old})^T \Gamma_{n-1}^{-1} y_{n-1}^{old}$$

residual error for best linear prediction for y in terms of x
 $y y - b y x^T \Sigma_x^{-1} b y x$

Error when predicting X_{n+1} based on last n observations

$$E \left(X_{n+1} - \hat{a}_n^T X_{n:1} \right)^2 = y(n) - (y_{n-1}^{old})^T \Gamma_n^{-1} y_{n-1}^{old}$$

$$= y(n) - \hat{a}_n^T y_{n-1}^{old}$$

$$= y(n) - \left(\hat{a}_{n-1}^{old} - a_n \Gamma_{n-1}^{-1} y_{n-1}^{old} \right) \begin{pmatrix} y_{n-1}^{old} \\ y(n) \end{pmatrix}$$

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$$= (y(0) - a_{n-1}^{old} x_{n-1}^{(1)}) - a_n (y_{n-1}^{(1)}) \Gamma_{n-1}^{-1} y_{n-1}^{(1)} - a_n x_{n-1}^{(1)}$$

$$= (y(0) - x_{n-1}^{(1)T} \Gamma_{n-1}^{-1} x_{n-1}^{(1)}) - a_n (x_{n-1}^{(1)T} \Gamma_{n-1}^{-1} x_{n-1}^{(1)})$$

$$\hat{a}_n [y(0) - x_{n-1}^{(1)T} \Gamma_{n-1}^{-1} x_{n-1}^{(1)}]$$

$$= (1 - a_n^2) (y(0) - x_{n-1}^{(1)T} \Gamma_{n-1}^{-1} x_{n-1}^{(1)})$$

$$= (1 - a_n^2) E (X_{n+1} - (a_{n-1}^{old})^T X_{n:2})^2$$

Error when predicting X_{n+1} based on last n observation

→ New error

$$E (X_{n+1} - a_n^T X_{n:2})^2$$

$$= (1 - a_n^2)$$

$|a_n| < 1$

⇒ error will reduce

$$E (X_{n+1} - (a_{n-1}^{old})^T X_{n:2})^2$$

→ old error

since a_n is a PACF

$$E (X_{n+1} - a_{n+1}^T X_{n:2})^2$$

$$= (1 - a_n^2) (1 - a_{n+1}^2)$$

$$< (1 - a_n^2) < 1$$

$$E (X_{n+1} - a_{n-1}^{old}^T X_{n:2})^2$$